

Lab07_Pan

```
library(DBI)
library(RSQLite)
```

Warning: pacote 'RSQLite' foi compilado no R versão 4.5.3

```
conn = dbConnect(SQLite(), 'C:/Users/artur/Documents/curso r/database.sqlite3')
```

Seleção de todos os professores que lecionam matérias de Estatística:

```
SELECT DISTINCT i.name as names_profs FROM subjects s
JOIN subject_memberships sm ON s.code=sm.subject_code
JOIN sections se ON se.course_offering_uuid=sm.course_offering_uuid
JOIN teachings t ON t.section_uuid=se.uuid
JOIN instructors i ON i.id=t.instructor_id
WHERE abbreviation = 'STAT'
```

Table 1: Displaying records 1 - 10

names_profs
MINJING TAO
DONALD PORTER
SHENG WANG
KUNLING HUANG
DONG XIA
JENNIFER NGUYEN
SEULKEE YUN
DUZHE WANG
YONGFENG WU
JARED BROWN

Seleção do professor mais difícil de passar (menor média):

```

SELECT
  i.name AS instructor_name,
  ROUND(
    SUM(
      gd.a_count * 4.0 +
      gd.ab_count * 3.5 +
      gd.b_count * 3.0 +
      gd.bc_count * 2.5 +
      gd.c_count * 2.0 +
      gd.d_count * 1.0 +
      gd.f_count * 0.0
    ) /
    NULLIF(SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_c
2
    ) AS media_notas
FROM subjects sub
JOIN subject_memberships sm ON sub.code = sm.subject_code
JOIN course_offerings co ON sm.course_offering_uuid = co.uuid
JOIN courses c ON co.course_uuid = c.uuid
JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
JOIN sections sec ON co.uuid = sec.course_offering_uuid AND gd.section_number = sec.number
JOIN teachings t ON sec.uuid = t.section_uuid
JOIN instructors i ON t.instructor_id = i.id
WHERE sub.abbreviation = 'STAT'
GROUP BY i.id, i.name, c.name
HAVING media_notas IS NOT NULL
ORDER BY media_notas
LIMIT 1

```

Table 2: 1 records

instructor_name	media_notas
NICHOLAS STEPHEN KEULER	2.51

Seleção do professor mais fácil de passar (maior média):

```

SELECT
  i.name AS instructor_name,
  ROUND(
    SUM(
      gd.a_count * 4.0 +

```

```

        gd.ab_count * 3.5 +
        gd.b_count * 3.0 +
        gd.bc_count * 2.5 +
        gd.c_count * 2.0 +
        gd.d_count * 1.0 +
        gd.f_count * 0.0
    ) /
    NULLIF(SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count),
    2
) AS media_notas
FROM subjects sub
JOIN subject_memberships sm ON sub.code = sm.subject_code
JOIN course_offerings co ON sm.course_offering_uuid = co.uuid
JOIN courses c ON co.course_uuid = c.uuid
JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
JOIN sections sec ON co.uuid = sec.course_offering_uuid AND gd.section_number = sec.number
JOIN teachings t ON sec.uuid = t.section_uuid
JOIN instructors i ON t.instructor_id = i.id
WHERE sub.abbreviation = 'STAT'
GROUP BY i.id, i.name, c.name
HAVING media_notas IS NOT NULL
ORDER BY media_notas DESC
LIMIT 1

```

Table 3: 1 records

instructor_name	media_notas
JUN SHAO	4

Seleção da matéria mais difícil de passar (menor média)

```

SELECT
    co.name AS subject_name,
    ROUND(
        SUM(
            gd.a_count * 4.0 +
            gd.ab_count * 3.5 +
            gd.b_count * 3.0 +
            gd.bc_count * 2.5 +
            gd.c_count * 2.0 +
            gd.d_count * 1.0 +

```

```

        gd.f_count * 0.0
    ) /
    NULLIF(SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_c
2
    ) AS media_notas
FROM subjects sub
JOIN subject_memberships sm ON sub.code = sm.subject_code
JOIN course_offerings co ON sm.course_offering_uuid = co.uuid
JOIN courses c ON co.course_uuid = c.uuid
JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
JOIN sections sec ON co.uuid = sec.course_offering_uuid AND gd.section_number = sec.number
JOIN teachings t ON sec.uuid = t.section_uuid
JOIN instructors i ON t.instructor_id = i.id
WHERE sub.abbreviation = 'STAT'
GROUP BY i.id, i.name, c.name
HAVING media_notas IS NOT NULL
ORDER BY media_notas
LIMIT 1

```

Table 4: 1 records

subject_name	media_notas
Intro Stats for Engineers	2.51

Seleção da matéria mais fácil (maior média):

```

SELECT
    co.name AS subject_name,
    ROUND(
        SUM(
            gd.a_count * 4.0 +
            gd.ab_count * 3.5 +
            gd.b_count * 3.0 +
            gd.bc_count * 2.5 +
            gd.c_count * 2.0 +
            gd.d_count * 1.0 +
            gd.f_count * 0.0
        ) /
        NULLIF(SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_c
2
    ) AS media_notas

```

```

FROM subjects sub
JOIN subject_memberships sm ON sub.code = sm.subject_code
JOIN course_offerings co ON sm.course_offering_uuid = co.uuid
JOIN courses c ON co.course_uuid = c.uuid
JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
JOIN sections sec ON co.uuid = sec.course_offering_uuid AND gd.section_number = sec.number
JOIN teachings t ON sec.uuid = t.section_uuid
JOIN instructors i ON t.instructor_id = i.id
WHERE sub.abbreviation = 'STAT'
GROUP BY i.id, i.name, c.name
HAVING media_notas IS NOT NULL
ORDER BY media_notas DESC
LIMIT 1

```

Table 5: 1 records

subject_name	media_notas
Sample Survey Theory&Method	4